Hugo Marquerie 11/05/2025

Teorema de Morera para rectángulos

Teorema 1 (de Morera para rectángulos).
Sea $\Omega \subset \mathbb{C}$ *abierto y* $f : \mathbb{C} \longrightarrow \mathbb{C}$ *continua en* Ω*, entonces*

$$\forall z \in \Omega : \exists D(z, r_z) \subset \Omega : \forall R \subset D(z, r_z) \text{ rectángulo} : \int_{\partial R} f = 0 \implies f \in \mathcal{H}(\Omega).$$

***Demostración*:** Sabemos que si $\forall f \in \mathcal{C}(D(z, r_z)) : \forall R \subset D(z, r_z) : \int_{\partial R} f = 0$, entonces $\exists F \in \mathcal{H}(D(z, r_z)) : F' = f$ en $D(z, r_z)$ (`teo-cauchy-goursat-convexo/`**??**).

Como $F \in \mathcal{H}(D(z, r_z))$, entonces $F' = f \in \mathcal{H}(D(z, r_z))$. ∎

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